Riemann Mapping Theorem

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

A complete authored walkthrough of riemann mapping theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

CA-05 / Complex Analysis / Report v1

I. TARGET AND IMPORT CONTRACT

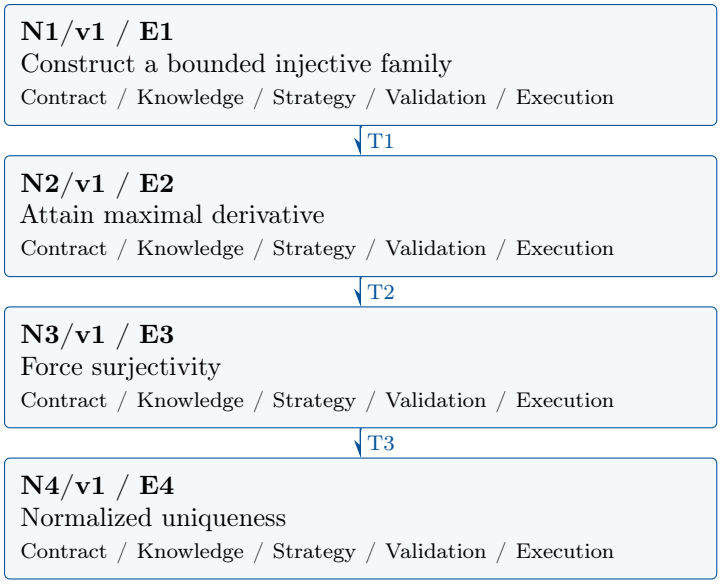
Theorem I.1. *For a nonempty simply connected proper domain $\Omega \subsetneq \mathbb{C}$ and $z_0 \in \Omega$, prove there is a unique biholomorphism $F : \Omega \to \mathbb{D}$ with $F(z_0) = 0$ and $F'(z_0) > 0$ real.*

A. Foundations and permitted imports

Use these imported complex-analytic modules: on a simply connected domain every nowhere-zero holomorphic function has a holomorphic logarithm and hence a square root; a locally bounded holomorphic family has a locally uniformly convergent subsequence (Montel); a locally uniform limit of injective holomorphic functions is injective or constant (Hurwitz consequence); the maximum modulus principle; Schwarz's lemma; Cauchy's derivative bound; disk automorphisms $\varphi_b(w) = (w-b)/(1-\mathrm{conj}(b)w)$. These supporting results are substantial and are not proved here. The target mapping theorem is not imported.

Also use the open mapping theorem and the local inverse theorem for a holomorphic injective map (its derivative is nonzero). These supporting complex-analysis modules are declared imports.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The family of injective maps $\Omega \to D$ with value 0 and positive derivative at z_0 is nonempty.

Receiving activity. N2: Attain maximal derivative

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Normalized family; Montel and Cauchy bounds; prior exports E1/v1

T2: N2/E2/v1 to N3. An injective normalized extremizer F with $F'(z_0) = \alpha > 0$.

Receiving activity. N3: Force surjectivity

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Extremizer F; hypothetical omitted $b \in D$; prior exports E2/v1

T3: N3/E3/v1 to N4. F is a biholomorphism $\Omega \to D$.

Receiving activity. N4: Normalized uniqueness

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Two biholomorphisms with the same normalization; prior exports E3/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Construct a bounded injective family

Object and role. Take a square root, omit an open disk, then reciprocate.

Inputs. Proper simply connected Ω and z_0; foundational imports stated in §1

C — Contract

Own E1/v1: The family of injective maps $\Omega \to D$ with value 0 and positive derivative at z_0 is nonempty.

K — Knowledge

Retain Proper simply connected Ω and z_0; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Take a square root, omit an open disk, then reciprocate.

V — Validation

Properness supplies a; a branch must be global on Ω.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Choose $a \notin \Omega$ and a branch $h(z) = \sqrt{z - a}$.

N1.2 It is injective since equality of h implies equality after squaring.

N1.3 Its image is disjoint from its negative: $h(z_1) = -h(z_2)$ would give $z_1 = z_2$ and $h(z_1) = 0$, impossible.

N1.4 Let $w_0 = h(z_0)$; the open image contains $B(w_0, r)$ for some $r > 0$, so it omits $B(-w_0, r)$.

N1.5 Thus $q(z) = 1/(h(z) + w_0)$ is injective, holomorphic, and bounded by 1/r.

N1.6 Scale it strictly into D, apply a disk automorphism to make the value at z_0 zero, and rotate so its nonzero derivative is positive real.

N1.7 This gives a nonempty family F of normalized injective maps $\Omega \to D$.

Export E1/v1. The family of injective maps $\Omega \to D$ with value 0 and positive derivative at z_0 is nonempty.

Acceptance gate. Properness supplies a; a branch must be global on Ω. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Properness supplies a; a branch must be global on Ω. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Attain maximal derivative

Object and role. Take an extremal sequence with a nonconstant limit.

Inputs. Normalized family; Montel and Cauchy bounds; prior exports E1/v1

C — Contract

Own E2/v1: An injective normalized extremizer F with $F'(z_0) = \alpha > 0$.

K — Knowledge

Retain Normalized family; Montel and Cauchy bounds; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Take an extremal sequence with a nonconstant limit.

V — Validation

A constant limit must be ruled out before invoking injective-limit preservation.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Choose a closed disk about z_0 contained in Ω.

N2.2 Cauchy's estimate bounds every family derivative at z_0 by one common finite constant.

N2.3 Let α be the supremum of the positive derivatives; it is positive by nonemptiness.

N2.4 Pick f_j with $f_j'(z_0) \to \alpha$.

N2.5 Montel provides a locally uniform limit F, with $F(z_0) = 0$ and $F'(z_0) = \alpha$ by Cauchy's formula.

N2.6 It is nonconstant.

N2.7 Hurwitz makes F injective, and the maximum modulus principle upgrades $|F| \leq 1$ to $|F| < 1$.

N2.8 Thus F belongs to the family and attains α.

Export E2/v1. An injective normalized extremizer F with $F'(z_0) = \alpha > 0$.

Acceptance gate. A constant limit must be ruled out before invoking injective-limit preservation. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: A constant limit must be ruled out before invoking injective-limit preservation. Recheck dependent nodes N3, N4.

C. N3/v1: Force surjectivity

Object and role. Construct a normalized map with a larger derivative.

Inputs. Extremizer F; hypothetical omitted $b \in D$; prior exports E2/v1

C — Contract

Own E3/v1: F is a biholomorphism $\Omega \to D$.

K — Knowledge

Retain Extremizer F; hypothetical omitted $b \in D$; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Construct a normalized map with a larger derivative.

V — Validation

The derivative improvement must be strict, not merely nondecreasing.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Suppose $b \in D \setminus F(\Omega)$.

N3.2 Because $F(z_0) = 0$, $b \neq 0$.

N3.3 Let $r = |b| \in (0, 1)$ and $H = \varphi_b \circ F$, which never vanishes.

N3.4 Take a holomorphic square root h of H on Ω.

N3.5 Since $|H| < 1$, h maps into D, and it is injective by squaring.

N3.6 Write $c = h(z_0)$; then $|c| = \sqrt{r}$.

N3.7 The normalized map $\psi_c \circ h$, followed by a rotation, belongs to the family.

N3.8 Its derivative modulus at z_0 is $\alpha(1 - r^2)/(2\sqrt{r(1} - r)) = \alpha(1 + r)/(2\sqrt{r}) > \alpha$ because $(1 - \sqrt{r})^2 > 0$.

N3.9 This contradicts maximality.

N3.10 Hence F is onto D.

N3.11 Its derivative never vanishes by holomorphic injectivity, and the local inverse theorem gives a holomorphic inverse.

Export E3/v1. F is a biholomorphism $\Omega \to D$.

Acceptance gate. The derivative improvement must be strict, not merely nondecreasing. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: The derivative improvement must be strict, not merely nondecreasing. Recheck dependent nodes N4.

D. N4/v1: Normalized uniqueness

Object and role. Apply Schwarz to the transition map and its inverse.

Inputs. Two biholomorphisms with the same normalization; prior exports E3/v1

C — Contract

Own E4/v1: Existence and uniqueness of the specified normalized biholomorphism.

K — Knowledge

Retain Two biholomorphisms with the same normalization; prior exports E3/v1; keep their assumptions and versions attached.

S — Strategy

Apply Schwarz to the transition map and its inverse.

V — Validation

Dropping the positive-derivative condition leaves rotational freedom.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 If F and G are normalized, $A = G \circ F^{-1}$ is an automorphism of D fixing 0.

N4.2 Schwarz applied to A and A^{-1} gives $|A'(0)| = 1$ and the equality case $A(w) = e^{i\theta}w$.

N4.3 But $A'(0) = G'(z_0)/F'(z_0)$ is positive real; hence it equals 1 and A is the identity.

N4.4 Thus $G = F$.

Export E4/v1. Existence and uniqueness of the specified normalized biholomorphism.

Acceptance gate. Dropping the positive-derivative condition leaves rotational freedom. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Dropping the positive-derivative condition leaves rotational freedom. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Construct a bounded injective family

1. Choose $a \notin \Omega$ and a branch $h(z) = \sqrt{z - a}$.
2. It is injective since equality of h implies equality after squaring.
3. Its image is disjoint from its negative: $h(z_1) = -h(z_2)$ would give $z_1 = z_2$ and $h(z_1) = 0$, impossible.
4. Let $w_0 = h(z_0)$; the open image contains $B(w_0, r)$ for some $r > 0$, so it omits $B(-w_0, r)$.
5. Thus $q(z) = 1/(h(z) + w_0)$ is injective, holomorphic, and bounded by 1/r.
6. Scale it strictly into D, apply a disk automorphism to make the value at z_0 zero, and rotate so its nonzero derivative is positive real.
7. This gives a nonempty family F of normalized injective maps $\Omega \to D$.

B. §4.N2 — Attain maximal derivative

1. Choose a closed disk about z_0 contained in Ω.
2. Cauchy's estimate bounds every family derivative at z_0 by one common finite constant.
3. Let α be the supremum of the positive derivatives; it is positive by nonemptiness.
4. Pick f_j with $f_j'(z_0) \to \alpha$.
5. Montel provides a locally uniform limit F, with $F(z_0) = 0$ and $F'(z_0) = \alpha$ by Cauchy's formula.
6. It is nonconstant.
7. Hurwitz makes F injective, and the maximum modulus principle upgrades $|F| \leq 1$ to $|F| < 1$.
8. Thus F belongs to the family and attains α.

C. §4.N3 — Force surjectivity

1. Suppose $b \in D \setminus F(\Omega)$.
2. Because $F(z_0) = 0$, $b \neq 0$.

3. Let $r = |b| \in (0, 1)$ and $H = \varphi_b \circ F$, which never vanishes.

4. Take a holomorphic square root h of H on Ω.

5. Since $|H| < 1$, h maps into D, and it is injective by squaring.

6. Write $c = h(z_0)$; then $|c| = \sqrt{r}$.

7. The normalized map $\psi_c \circ h$, followed by a rotation, belongs to the family.

8. Its derivative modulus at z_0 is $\alpha(1 - r^2)/(2\sqrt{r(1} - r)) = \alpha(1 + r)/(2\sqrt{r) > \alpha}$ because $(1 - \sqrt{r)^2 > 0}$.

9. This contradicts maximality.

10. Hence F is onto D.

11. Its derivative never vanishes by holomorphic injectivity, and the local inverse theorem gives a holomorphic inverse.

D. §4.N4 — Normalized uniqueness

1. If F and G are normalized, $A = G \circ F^{-1}$ is an automorphism of D fixing 0.

2. Schwarz applied to A and A^{-1} gives $|A'(0)| = 1$ and the equality case $A(w) = e^{i\theta}w$.

3. But $A'(0) = G'(z_0)/F'(z_0)$ is positive real; hence it equals 1 and A is the identity.

4. Thus $G = F$.

□

V. WORKED ORCHESTRATION

For the upper half-plane Ω and $z_0 = i$, $F(z) = i(z - i)/(z + i)$ maps Ω onto D. The Cayley transform has unit-circle boundary values and maps i to 0; multiplication by i rotates the disk. Direct differentiation gives $F'(i) = 1/2 > 0$, matching the normalization.

A. Boundary and hypothesis audit

A multiply connected annulus cannot be biholomorphic to D because biholomorphisms preserve fundamental groups. For $\Omega = C$, a map into D would be bounded entire and constant by Liouville. Validation checks these domain exclusions before attempting the square-root/extremal proof.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A multiply connected annulus cannot be biholomorphic to D because biholomorphisms preserve fundamental groups. For $\Omega = C$, a map into D would be bounded entire and constant by Liouville. Validation checks these domain exclusions before attempting the square-root/extremal proof.

C. Version and recheck

Version changed contracts and recheck the revised export and every dependent node. Diagnostic feedback is not a proof premise.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Harvard complex analysis: Cauchy formula, argument principle, and Riemann mapping lectures.

- Harvard Riemann Mapping Theorem lecture.